

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2023-P2-Model)** **Date: 04-02-2026**

Time: 02.00Pm to 05.00Pm **GTA-14** **Max. Marks: 180**

04-02-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2023-P2)_GTA-14_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Name of the Student: _____ **H.T. NO:** | | | | | | | | |

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JEE-ADVANCE-2023-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with Integer Answer Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Paragraph NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 25 – 30)	Questions with Integer Answer Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Paragraph NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 37)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 38 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 42 – 47)	Questions with Integer Answer Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Paragraph NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

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MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

1. ABC and BDE are two equilateral triangles such that D is the mid-point of BC, AE intersect BC at F. *If* $2|z-z_d| = |z_a-z_c|$ and

$\left(\frac{|z-z_a|_{\max}}{|z-z_a|_{\min}}-2\right)^2 = k\frac{area(\Delta FED)}{area(\Delta AFC)}$, then value of k is equal to _____ {where z represents complex number, z_a, z_d & z_c are the affixes of A, D & C respectively}

A) 21 **B)** 22 **C)** 23 **D)** 24

2. If 'P' be a point on unit circle with centre at point 'O'. If Q is another point such that 'OQ' inclined at angle $\frac{\pi}{3}$ **to** 'OP' and $|\overrightarrow{OT}| = |\overrightarrow{OQ}| = r$, then the interval in which 'r' lies, so that $2|\overrightarrow{AT}| = |\overrightarrow{PQ}|$, and $\overrightarrow{OA} = \hat{i}$, is.

A) $\left[\frac{7+\sqrt{13}}{6}, \infty\right)$ **B)** $\left[\frac{7-\sqrt{13}}{6}, \frac{7+\sqrt{13}}{6}\right]$

C) $\left(-\infty, \frac{7-\sqrt{13}}{6}\right]$ **D)** $\left(\frac{7-\sqrt{13}}{6}, \frac{17+\sqrt{13}}{6}\right)$

3. A student is trying to find the number of points of non-differentiability of a function

$P(x) = |x-1|\cos|x-2|\cdot\sin|x-1| + (x-3)|x^2-5x+4|, x \in \mathbb{R}$

He wishes to solve the problem in as much less time as possible. Eventually he arrived at the following list of observations.

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1) For the function $y=f(x)g(x)$ if both functions $f(x),g(x)$ are not differentiable at $x=a$ but $f(a)=0$ and $g(a)=0$ then y is differentiable at $x=a$.

2) The functions $f(x)=|x|$ is not differentiable at $x=0, g(x)=\sin|x|$ is not differentiable at $x=0$ and $f(0)=g(0)=0$. The function $|x|\sin|x|$ is differentiable at $x=0$.

3) If $y=f(x).g(x)$ be a function and the $f(x)$ is not differentiable at $x=a$ and function $g(x)$ is such that $g(a)\neq 0$ then y is not differentiable $x=a$

4) The function $f(x)=|x^2-6x+8|$ is not differentiable at $x=4$ and $g(x)=|x-2|$ is such that $g(4)\neq 0$. The function $|x^2-6x+8|\cdot|x-2|$ is not differentiable at $x=4$.

5) Let $y=f(x)\cdot g(x)$ be a function and the function $f(x)$ is not differentiable at $x=a$ and function $g(x)$ is such that $g(a)=0$ then y is differentiable at $x=a$

6) The function $f(x)=|x^2-6x+8|$ is not differentiable at $x=2$ and $g(x)=|x-2|$ is such that $g(2)=0$. The function $|x^2-6x+8|\cdot|x-2|$ is differentiable at $x=2$

7) The number of points of non-differentiability of $P(x)$ is 2.

8) If both the functions $f(x)$ and $g(x)$ are not monotone on $\mathbb{R}$, then their sum $f(x)+g(x)$ is not monotone on $\mathbb{R}$.

9) If a function $f(x)$ is continuous and increasing at the point $x=a$, then there is a neighbourhood $(a-\delta, a+\delta), \delta>0$ where the function is also increasing.

The number of statements, which are incorrect in the above list is

A) 3 **B)** 2 **C)** 5 **D)** 0

4. If A and B are orthogonal matrices of order 2 such that ABA^{-1} is a symmetric matrix, where A, B and $ABA^{-1}\neq I, -I$. Then which of the following statements are incorrect? (Note: $|X|$, $tr(X)$ and $adj(X)$ denote determinant value, trace and adjoint of matrix X respectively)

A) $|B|$ is equal to -1 **B)** $tr\left(\left(ABA^{-1}\right)^{2017}\right)$ is equal to 0

C) $tr\left(B^2\right)$ is equal to 2 **D)** Sum of elements of $\left(\frac{adj(B)}{\sqrt{2}}\right)^2$ is equal to 2

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

5. Any complex number in the polar form can be an expression in Euler's form as $\cos\theta + i\sin\theta = e^{i\theta}$. This form of the complex number is useful in finding the sum of the series $\sum_{r=0}^{n} {}^{n}C_r\left(\cos\theta + i\sin\theta\right)^r = \sum_{r=0}^{n} {}^{n}C_r\left(\cos r\theta + i\sin r\theta\right)$

$= \sum_{r=0}^{n} {}^{n}C_r e^{ir\theta} == \sum_{r=0}^{n} {}^{n}C_r\left(e^{i\theta}\right)^r = \left(1+e^{i\theta}\right)^n.$

Also, we know that the sum of binomial series does not change, if r is replaced by $(n-r)$. Using these facts, choose the correct option.

A) $\sum_{r=0}^{100} {}^{100}C_r\left(\sin rx\right) = 2^{100}\cos^{100}\left(\frac{x}{2}\right).\sin 50x$

B) In ΔABC, $\sum_{r=0}^{50} {}^{50}C_r\, a^r\, b^{50-r}\cos\left[rB-(50-r)A\right] = (2s-a-b)^{50}$,
Where a, b and c have usual meanings, 2s is perimeter of a triangle.

C) $f(\pi/8) = 1$, if $f(x) = \dfrac{\sum_{r=0}^{50} {}^{50}C_r.\sin^2 rx}{\sum_{r=0}^{50} {}^{50}C_r.\cos^2 rx}$

D) Only (a) and (c) are true

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6. In general, If the entries of a matrix depend in a Differentiable manner on a parameter 'x'

$$\begin{bmatrix} a_{11}(x) & a_{12}(x) & a_{13}(x)..... & a_{1n}(x) \\ a_{21}(x) & a_{22}(x) & a_{23}(x)..... & a_{2n}(x) \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1}(x) & a_{n2}(x) & a_{n3}(x)..... & a_{nn}(x) \end{bmatrix}$$

. Then the determinant is a differentiable function of 'x' and its derivative equal to

$$\begin{vmatrix} a'_{11}(x) & a'_{12}(x) & a'_{13}(x)..... & a'_{1n}(x) \\ a_{21}(x) & a_{22}(x) & a_{23}(x)..... & a_{2n}(x) \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1}(x) & a_{n2}(x) & a_{n3}(x)..... & a_{nn}(x) \end{vmatrix} + \begin{vmatrix} a_{11}(x) & a_{12}(x) & a_{13}(x)..... & a_{1n}(x) \\ a'_{21}(x) & a'_{22}(x) & a'_{23}(x)..... & a'_{2n}(x) \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1}(x) & a_{n2}(x) & a_{n3}(x)..... & a_{nn}(x) \end{vmatrix} + ...$$

This follows by applying the product role to the following determinant function

$$f(x) = \begin{vmatrix} x+a_1 & x & ..... & x \\ x & x+a_2..... & & x \\ \vdots & \vdots & & \vdots \\ x & x & .....x+a_n & \end{vmatrix}$$

A) represents a linear function and $f(x) = (a_1 a_2 .... a_n)\left[1+\left(\frac{1}{a_1}+\frac{1}{a_2}+....\frac{1}{a_n}\right)x^2\right]$

B) If $x=1$, then $f(1) = \begin{vmatrix} 1+a_1 & 1 & ..... & 1 \\ 1 & 1+a_2..... & & 1 \\ \vdots & \vdots & & \vdots \\ 1 & 1 & .....1+a_n & \end{vmatrix} = (a_1 a_2 .... a_n)\left[1+\left(\frac{1}{a_1}+\frac{1}{a_2}+...\frac{1}{a_n}\right)\right]$

C) If $a_1 = 1, a_2 = 2, a_3 = 3, ..., a_{2026} = 2026$. Then

$$f(x) = (1.2.3.4.....2026)\left[1+\left(\frac{1}{1}+\frac{1}{2}+....\frac{1}{2026}\right)x\right]$$

D) If $a_1 = 1, a_2 = 2, ..., a_{2026} = 2026$, and $x = 1$, then $f(1) = (2026)!\left[1+\left(\frac{1}{1}+\frac{1}{2}+...\frac{1}{2026}\right)\right]$

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7. Consider the following frequency distribution:

Value	4	5	8	9	6	12	11
Frequency	5	f_1	f_2	2	1	1	3

Suppose that the sum of the frequencies is 19 and the median of this frequency distribution is 6.

For the given frequency distribution, let α denote the mean deviation about the mean, β denotes the mean deviation about the median, and σ^2 denotes the variance.

Then which of the following statement(s) is/are correct?

A) $7f_1+9f_2$ is equal to 55 **B)** 19α is equal to 48

C) 19β is equal to 47 **D)** $19\sigma^2$ is equal to 146

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

8. The number of all the pairs of natural numbers (x, y) which satisfy the equation $2x(xy-2y-3)=(x+y)(3x+y)$ is

9. Let p, q, r and s are the probabilities of raining at four different places at some fixed moment, if M and N are the maximum and minimum value of $3(p^2+q^2+r^2+s^2)-2(p+q+r+s)+4$, then value of $[M-N]$ is (where $[\bullet]$ represent greatest integer function)

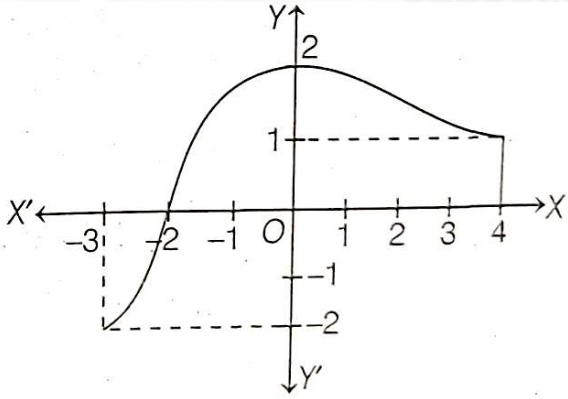
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10. Consider a function f whose domain is $[-3, 4]$ and range is $[-2, 2]$ with following graph (see the figure). Also consider domain and range of $\phi(x)=f(|x|)$ are $[\alpha, \beta]$ and $[\gamma, \delta]$ respectively, such that $(\beta-\alpha+\gamma+\delta)=\lambda$ as well as

$\Psi(x)=\left|f(x)-\frac{3}{2}\right|$ has range $[p, q]$ and n is the number of real solutions of $\Psi(x)=\frac{1}{4}$ such that $(n+p+2q)=\mu$.

If $k=\frac{(\textit{Difference of units \& ten's place digits in the value of } \lambda)}{(\textit{sum of digits in the value of } \mu)}$ then $[2\sin k+\tan(2k)]$ is equal to _____



11. A function $f(x)$ is defined for $x \in [-1, 1]$ by

$$f(x)=1+x+\frac{1}{2}x^2+\frac{1}{2}x^3+\frac{1.3}{2.4}x^4+\frac{1.3}{2.4}x^5+\frac{1.3.5}{2.4.6}x^6+\frac{1.3.5}{2.4.6}x^7+----$$

Then the value of $f\left(\frac{15}{17}\right)$ is equal to _____

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NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
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12. If $\sum_{k=1}^{n}\left(k^2+5k+7\right)(k+2)!+3.(3!)=2029\,(2029!)$

Then the value of n is equal to _____

13. Given, $m, n \in \{0, 1, 2, 3, 4, ...,19, 20\}$. Consider the system of equations

$x + y + z = 4$

$2x + y + 3z = 6$

$x + 2y + mz = n$

Let P: denotes the number of ordered pairs $(m,\ n)$. So that the system of equations has unique solution.

Q: Denotes the number of ordered pairs $(m,\ n)$. So that the system of equation has no solution.

R: Denotes the number of ordered pairs $(m,\ n)$. So that the system of equation has infinite solutions.

The value of $\frac{P}{Q.R}$ is

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

Three numbers are chosen at random, one after another with replacement, from the set $S = \{1, 2, 3,..., 100\}$. Let p_1 be the probability that the maximum of chosen numbers is at least 81 and p_2 be the probability that the minimum of chosen number is at most 40.

14. The value of $\frac{625}{4}\,p_1$ is _____

15. The value of $\frac{125}{4}\,p_2$ is _____

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PARAGRAPH-II

A function $f(x):\mathbb{R}\to\mathbb{R}$ *is such that* $f(x)$ is continuous in and has its first derivative on the interval $[0,1]$ and satisfies $f(0)=\frac{6-4\sqrt{2}}{3}$, $f(1)=2$ and $f'(x)>0\ \forall\ x\in[0,1]$ the function is such that the minimum value of the integral

$I=\int_0^1\sqrt{2+2\sqrt{2x-x^2}+\left[f'(x)\right]^2}\,dx$ say the minimum

Without the limits can be expressed as

$\lambda\left\{-\frac{2}{3}\left[g(2-x)\right]^3+\frac{2}{3}\left[g(x)\right]^3+f(x)\right\}$ where λ is constant

16. Define $h(x)=2\sqrt{6}\ g(x)$. If a tangent to the curve $h(x)$ meet the curve $xy=2$ at points A and B. Then the midpoints of such line segments AB lie on a Parabola with directrix x = _____

17. If the minimum value of the integral is equal to α, and the value of $f(2)$ is equal to 'β', Then the value of $3\alpha+3\beta-4\sqrt{2}$ is _____

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PHYSICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

18. A very long straight conducting wire, lying along the Z axis carries a current of 2A. The integral $\int \bar{B}.d\bar{l}$ is computed along the straight line PQ where P has the co-ordinates (2cm, 0, 0) and Q has the co-ordinates (2cm, 2cm, 0). The integral has the magnitude (in SI units)

A) zero **B)** $8\pi\times10^{-7}$ **C)** $2\pi\times10^{-7}$ **D)** $\pi\times10^{-7}$

19. When you continuously over-write the symbol '8'at the same place, on the page it is like the tip of the pen is executing two simultaneous oscillations, one along the length of the page $(\uparrow)$ and the other along the width $(\rightarrow)$ of the page. The ratio of the respective angular frequencies is

A) 1 : 1 **B)** 1 : 2 **C)** 2 : 1 **D)** $\sqrt{2}:1$

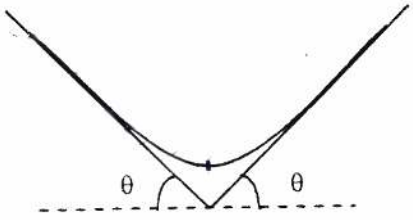
20. A spherical soap bubble (surface tension=*S*) encloses *n* moles of a monoatomic ideal gas. The gas is heated slowly so that the surface area of the bubble increases by $\frac{3nR}{2S}$ per unit increment in temperature of the gas. The specific heat for this process is (R = universal gas constant)

A) $\frac{3R}{2}$ **B)** $\frac{5R}{2}$ **C)** $\frac{7R}{2}$ **D)** $3R$

21. A rope rests on two platforms which are both inclined at an angle θ, as shown. The rope has uniform mass density, and its coefficient of friction with the platforms is 1. The system has left – right symmetry. The rope is arranged such that the largest possible fraction of the rope does not touch the platforms. The maximum value of θ which allows this is

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A) $\frac{\pi}{2}$ B) $\frac{\pi}{4}$ C) $\frac{\pi}{8}$ D) $\frac{\pi}{6}$

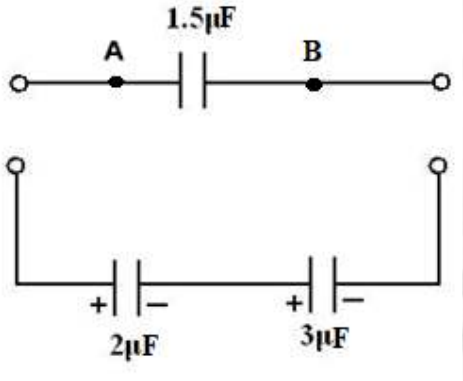
SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
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Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

22. Two capacitors of $2\mu F$ and $3\mu F$ are charged to 150V and 120 V respectively. The plates of capacitors are connected as shown in figure. An uncharged capacitor of capacity $1.5\mu F$ falls onto the free end of the wire and sticks to it to complete the circuit. Then



A) Final charge on $1.5\mu F$ capacitor is $180\mu C$

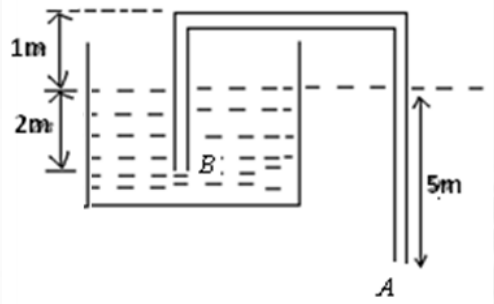
B) Final charge on $2\mu F$ capacitor is $120\mu C$

C) Positive charge flows through A from right to left

D) Positive charge flows through B from left to right

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23. In the figure shown of a siphon, the oil (density = $2gm/cm^3$) is initially present in the entire length of the siphon with end A closed. Ignore viscosity and surface tension effect. Take $g = 10m/s^2$, $P_{atm} = 10^5 N/m^2$.Now, when A is opened, during the subsequent period,(Assume the initial level of liquid in the vessel is maintained)



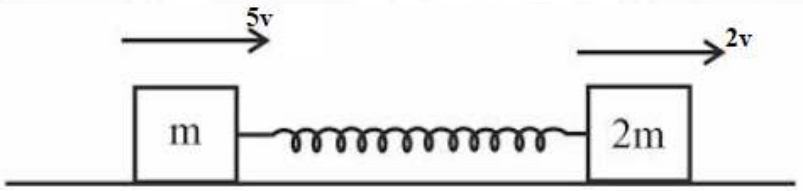
A) Liquid continues to come out with a speed $10m/s$

B) Liquid continues to come out with a speed $2\sqrt{15}m/s$

C) Liquid continues to come out with a speed $2\sqrt{30}m/s$

D) Speed of oil inside siphon tube is same at all points.

24. Two blocks of mass m and 2m are connected by a massless spring of spring constant k and are moving on a smooth horizontal floor. At an instant t=0, block of mass m has velocity '$5v$' towards right (along the spring) and block of mass 2m has velocity '$2v$' in the same direction. At t=0, the spring is in undeformed state. During the subsequent motion of the system



A) Maximum change in potential energy Stored in the spring from t=0 is $3mv^2$.

B) Maximum speed of block of mass 2m is $4v$

C) Minimum speed of block of mass m is v

D) Maximum relative speed of the two blocks is $3v$

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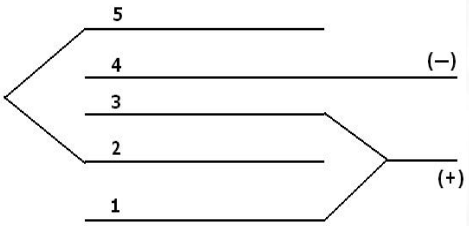
SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases..

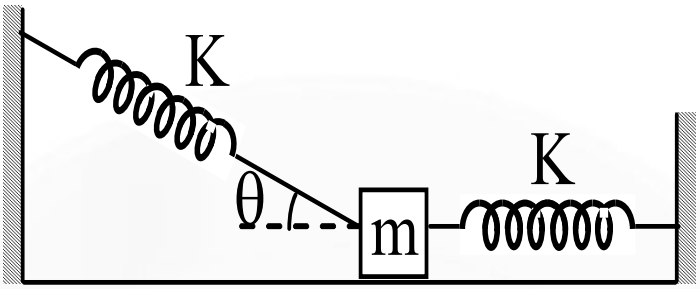
25. The electric field strength depends on the x, y, z coordinates according to the law $E = \dfrac{a\left(x\bar{i} + y\bar{j} + z\bar{k}\right)}{\left(x^2 + y^2 + z^2\right)^{3/2}}$ where $a = 280 Nm^2c^{-1}$. If the potential difference between the points $(3,2,6)$ and $(0,3,4)$ is $x^2\, volt$, find the value of x (Coordinates mentioned are in meter)

26. Five identical conducting plates 1, 2, 3, 4 and 5 are fixed parallel plates equidistant from each other with plate area A each and the distance between successive plates is d. A conductor connects plates 2 and 5 while another conductor joins 1 and 3. The junction of 1 and 3 and plate 4 are connected across a DC source. If the effective capacity of the system is $\frac{x}{y}.\frac{\in_0 A}{d}$ find $x+y$ where x and y are smallest posible integers.



27. A block of mass m placed on a smooth horizontal floor is connected with two light springs of spring constant K. The angular frequency of small horizontal oscillations of the block along horizontal plane is $n\ rad\ s^{-1}$. Find n. Given: $m = 41\ kg$, $K = 100\ N/m$, $\theta = 37^\circ$.(Initially springs shown are in undeformed state)

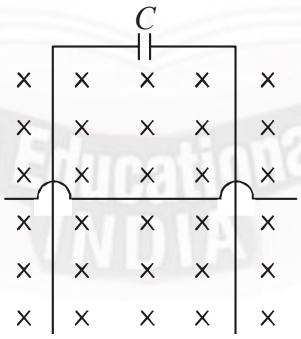
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28. A boy runs on a circular path of radius $R = 28m$ with a constant speed $u = 4m/s$. Another boy starts from the centre of the path to catch the first boy. The second boy always remains on the radius connecting the centre of the circle and the first boy and maintains magnitude of his velocity a constant $V = 4m/s$ .If the time of chase is (10+x) sec then x= -----$\left(\pi = \frac{22}{7}\right)$

29. A piece of ice (specific heat $= 0.5 cal/gm/^0C$ and latent heat $= 80 cal/gm$) of mass m gram is at -5^0C at atmospheric pressure. It is given 420 J of heat so that the ice starts melting. Finally when the ice-water mixture is in equilibrium, it is found that 1 gm of ice has melted. Assuming, there is no other heat exchange in the process, the value of m in gram is $(J = 4.2 J/cal)$

30. Two metal bars are fixed vertically and are connected on the top by a capacitor of capacitance C. A sliding conductor of length l and mass m slides with its ends in contact with the bars. The arrangement is placed in a uniform horizontal magnetic field directed normal to the plane of the figure. The conductor is released from rest. If $CB^2l^2 = 4m$ then the displacement of the conductor (in m) at 2s after release is _____.
[Assume g = 10 m/s^2]



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SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks:* +3** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

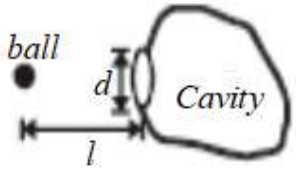
The Young's double-slit experiment is done in a medium of refractive index 4/3. A light of 600 nm wavelength is falling on the slits having 0.45 mm separation. The upper slit s_1 is covered by a thin glass sheet of thickness $10.4\mu m$ and refractive index 1.5. The interference pattern is observed on a screen placed 1.5m from the slits. All the wavelengths given in this problem are for the given medium of refractive index 4/3. Ignore dispersion (Assume that source is symmetric to the slits, centre of the screen is taken as origin)

31. The location of the zeroth order maximum (bright fringe) with zero path difference on the y-axis (in mm) is

32. Now, if 600nm light is replaced by light of wave length range 400nm to 700 nm. Find difference in wavelengths (in A^0) of the light that form maxima exactly at centre point of screen.

PARAGRAPH-II

A small spherical ball of radius r is maintained at temperature T. The emissivity of the surface of the ball is 1. There is a cavity with a small hole of diameter d and perfectly reflecting outer surfaces kept at a large distance l from the center of the ball.



33. The rate of amount of radiation (emitted from the ball) falling on the hole is approximately equal to $\frac{\pi d^2 r^2 \sigma T^4}{nl^2}$. The value of n is

34. The equilibrium temperature of the cavity at which the amount of power striking the hole from both sides is equal to $T^a\sqrt{\frac{r}{l^b}}$. The value of a – b is

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CHEMISTRY **Max Marks: 60**

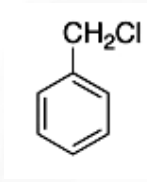
SECTION–I
(SINGLE CORRECT ANSWER TYPE)

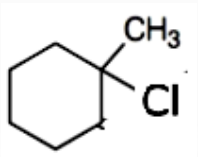
This section contains **FOUR (04)** questions.

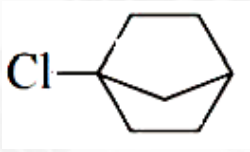
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

35. Which of the following can give a ppt of AgCl with both aqueous $AgNO_3$ and alcoholic $AgNO_3$

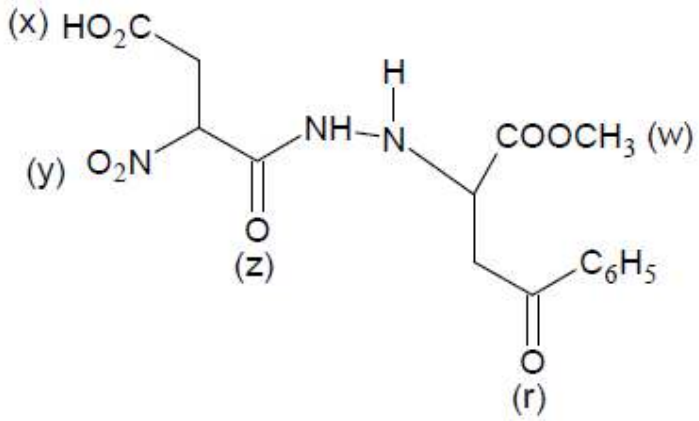
1. 

2. 

3. $(CH_3)_3CCl$ 4. 

A) 1,2,3,4 **B)** 1,2,3 **C)** 1,2 **D)** 2,3,4

36. Identify the correct statement(s) given for the reduction of the following compounds



1. $LiAlH_4$ reduces x, y, z, w, r
2. $NaBH_4$ reduces 'r' only
3. Na / C_2H_5OH reduces x, r and w only
4. One equivalent of B_2H_6 / THF can only reduce x

A) 1,2,3,4 **B)** 1,2,3 **C)** 1,2,4 **D)** 2,3,4

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37. Consider the following statements:

(i) Low oxidation states are found when a complex compound has ligands capable of sigma-acceptor character in addition to the π-bonding

(ii) The magnetic properties of the actinoids are more complex than those of the lanthanoids

(iii) $Cu^{+2}_{(aq)}$ is more stable than $Cu^{+1}_{(aq)}$

(iv) The highest Mn fluoride is MnF_7 whereas the highest oxide is Mn_2O_7 .

(v) Interstitial compounds have high melting points, they are very hard, and are chemically inert

(vi)The ions Ti^{2+}, V^{2+} and Cr^{2+} are strong reducing agents and will liberate hydrogen from a dilute acid

(vii) The chromate ion is tetrahedral whereas the dichromate ion consists of two tetrahedra sharing one corner with Cr–O–Cr bond angle of 126°.

(viii) The 3d series element with lowest enthalpy of atomization is zinc

(ix) The order of E^o values for M^{+3}/M^{+2} for Co, Mn and Fe follows the order Co>Mn>Fe

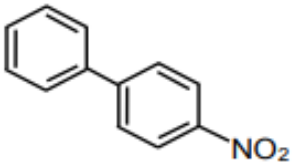
Of the above statements, the ratio of number of correct statements to the number of incorrect statements is:

A) 2 **B)**1.25 **C)**3.5 **D)**3

38. Which of the following is correct

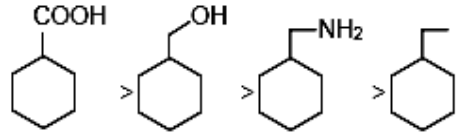
1) Number of different types of dipepitdes produced using a mixture of glycine and L-valine is 4 and the number of optically active dipeptides formed in this mixture will be 3

2) One of the product obtained on reaction of biphenyl (Ph-Ph) with nitrating mixture $(HNO_3 + H_2SO_4)$ is



3) pKa value of picric acid is more than that of formic acid

4)



(order of boiling point)

A)1,2,3,4 **B)**1,2,4 **C)**1,2,3 **D)**2,3,4

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

39. A quantity of 6 g $NaOH$ and 4.4 g CO_2 is allowed to react to form Na_2CO_3 or $NaHCO_3$ or both. Which of the following is correct statement regarding the reaction?

A) NaOH is the limiting reagent if there is no any formation of $NaHCO_3$.

B) NaOH is the limiting reagent if there is no any formation of Na_2CO_3.

C) Equal masses of Na_2CO_3 and $NaHCO_3$ are formed if none of the reactant is left.

D) The total mass of reaction mixture will be 10.4 g after the end of reaction, in any possible case.

40. In qualitative inorganic analysis of a water-soluble salt mixture (salt AB + salt XY) both the cations were identified as sulphides. In the tests for anions sodium carbonate extract when treated with $AgNO_3$ gave yellowish precipitate soluble with difficulty in NH_4OH while the other anion can be confirmed with brown ring test.

(K_{SP} for AS = 1×10^{-44} and XS = 1.4×10^{-24})

Idenfity the CORRECT statement about the analysis.

A) H_2S can be used under appropriate conditions of pH to separate and identify the cations.

B) Cation 'A' will be precipitated under acidic condition as the concentration of sulphides ions required is low.

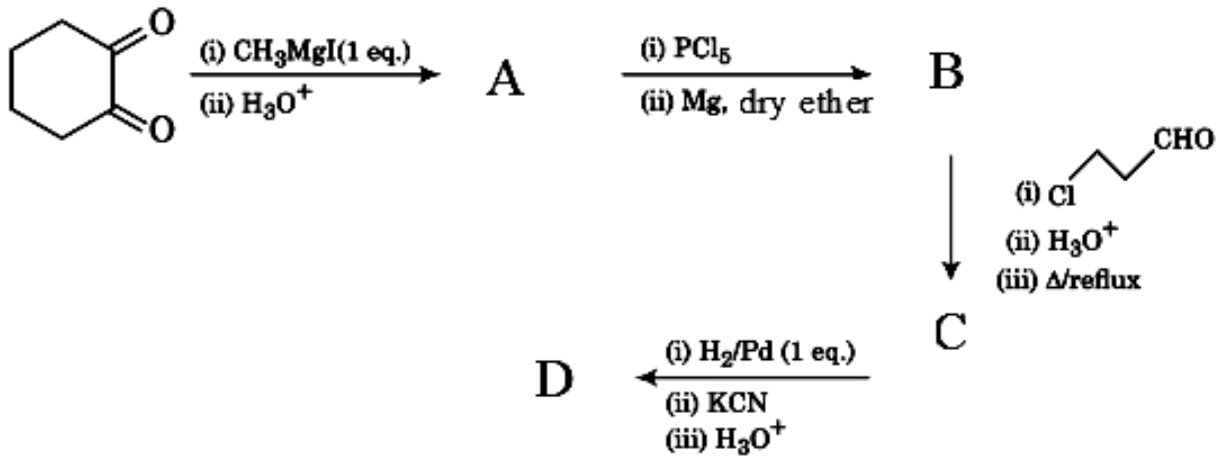
C) The anions are NO_3^- and Cl^-.

D) Cation X will be precipitated as sulphides under alkaline condition, as the concentration of sulphides ions required is very high.

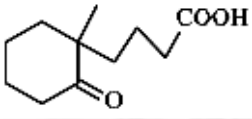
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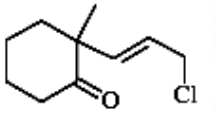

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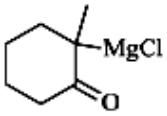
41.

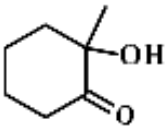


Identify the correct statements on the basis of above reaction sequence

A) Product D is 

B) Product C is 

C) Product B is 

D) Product A is 

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

42. The overall formation constant of $\left[Co\left(NH_3\right)_6\right]^{2+}$ in aqueous solution is 10^5 and the standard potential for reduction of $Co^{3+}\left(aq\right)$ and $\left[Co\left(NH_3\right)_6\right]^{3+}\left(aq\right)$ are as follows

$$Co^{3+}\left(aq\right)+e \rightleftarrows Co^{2+}\left(aq\right)\ E^0=1.90$$

$$\left[Co\left(NH_3\right)_6\right]^{3+}\left(aq\right)+e \rightleftarrows \left[Co\left(NH_3\right)_6\right]^{2+}\left(aq\right)\ ,E^0=+\ 0.1\,V$$

If the nearest overall formation constant of the $\left[Co\left(NH_3\right)_6\right]^{3+}$ ion is 10^x. Then the value of x would be $\left(Given: E_{cell}=E^0_{cell}-0.06\log Q\right)$

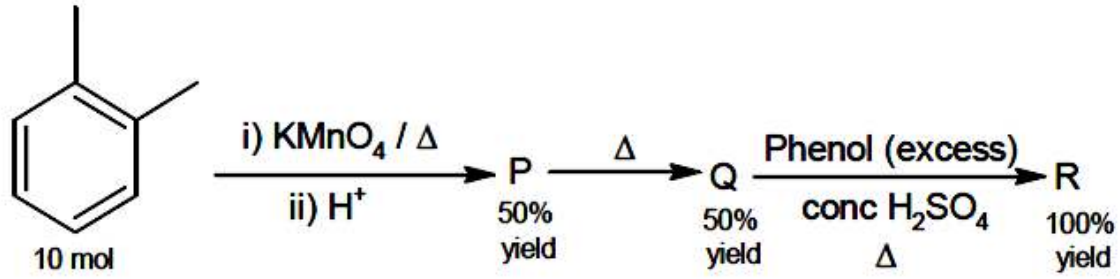
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43. A quantity of 5.0 kg liquid water is poured in an evacuated room of capacity $40\,m^3$ maintained at 300 K. Assume ideal gas behavior for water vapour. The vapour pressure of water at 300 K is 0.04 atm (R = 0.08 L – atm/K – mol).The percentage of liquid water that will vaporize is?

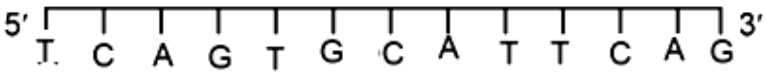
44. If P, Q, R, S is the total number of unpaired electrons in t_{2g} or t_2 set of orbitals in $K_3\left[Fe(CN)_6\right]$,$\left[FeCl_4\right]^{2-}$,$\left[Fe(H_2O)_6\right]Cl_3$,$\left[FeF_6\right]^{4-}$ respectively, then what is the value of P + Q + R + S?

45. A gaseous mixture contains hydrogen atoms in the 4th excited state, He^+ ions in 3rd excited state and Li^{2+} in 2nd excited state. The number of spectral lines of different wave lengths obtained in the emission spectrum of this sample when all these atoms/ions return to the ground state is

46. The amount of R produced in grams in the following reaction is



47. For a double strand DNA, one strand is given below



The amount of energy required to split the double strand DNA into two single strands is _____ kcal/mol

[Given: Average energy per H-bond for A-T base pair = 1.0 kcal/mol , G-C base pair = 1.5 kcal/mol, and A-U base pair =1.25kcal/mol. Ignore electrostatic repulsion between the phosphate groups.]

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SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

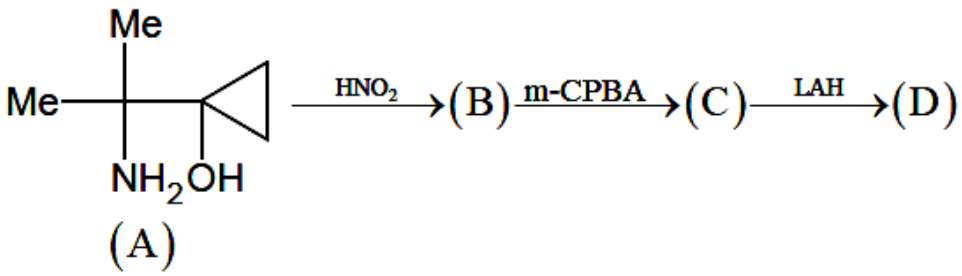
Boron trifluoride when heated with sodium hydride at 450 K, a highly toxic gas 'P' and a salt 'Q' are formed. Compound 'P' on hydrolysis produces compound 'R' and $H_2(g)$. Compound 'P' on complete combustion produces an acidic oxide 'M' and a neutral oxide 'N'.

[Given atomic mass: H = 1, B = 11, Na = 23, F = 19, O = 16]

48. The mass in gram of compound 'M' produced by the complete combustion of 5.52g of compound 'P' is_____

49. The molecular weight of R in grams is___

PARAGRAPH-II



[Given atomic mass: H = 1, C = 12, O = 16, N = 14]

50. The Molecular Formula of compound (B) is $C_xH_yO_z$. Then the value of x + y + z = ____

51. The Molecular mass (gm/mole) of compound (D) is:

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VANGALA AJAY REDDY
APP. NO. 250302255592
CLASSROOM STUDENT FROM GRADE IX - XII



DEVDUTTA MAJHI
APP. NO. 250310016185
DLP/AITS STUDENT



TOSHNIWAL SHIVEN
APP. NO. 250310391420
DLP/AITS STUDENT



SAKSHAM JINDAL
APP. NO. 250310236696
DLP/AITS STUDENT

BELOW 100	BELOW 500	BELOW 10	BELOW 100	BELOW 1000
31	95	10	98	579
ALL INDIA OPEN CATEGORY RANKS	ALL INDIA OPEN CATEGORY RANKS	ALL INDIA CATEGORY RANKS COUNT	ALL INDIA CATEGORY RANKS COUNT	ALL INDIA CATEGORY RANKS COUNT

TOTAL QUALIFIED RANKS FOR JEE ADVANCED-2025 22,094

*DLP/AITS

JEE 2025 STARS SHINE BRIGHT

Sri Chaitanya Tops JEE ADVANCED

ALL INDIA OPEN CATEGORY RANKS



RUTVIK SAI
H.T.No. 256055278 (OBC-NCL)



MAJID MUJAHID HUSAIN
H.T.No. 251134112*



UJJWAL KESARI
H.T.No. 252016104*



AKSHAT KUMAR CHAURASIA
H.T.No. 254065055*

BELOW 100	BELOW 500	BELOW 1000	BELOW 1000	NUMBER OF QUALIFIED RANKS
ALL INDIA OPEN CATEGORY RANKS	ALL INDIA OPEN CATEGORY RANKS	ALL INDIA OPEN CATEGORY RANKS	ALL INDIA CATEGORY RANKS COUNT	
29	113	205	745	4,212

*DLP/AITS

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